

D906

Methods of Teaching Secondary Mathematics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher planning on a Sunday evening drafts goals for a lesson on slope. Her district uses a new online gradebook. Which is the best learning goal?

- A. Students will complete the slope worksheet.
- B. Students will learn everything about linear functions.
- C. Students will explain slope as a constant rate of change and identify it in a table, a graph, and an equation.
- D. Students will enjoy working with slope.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A teacher with 110 students across four sections is choosing an area task. The department just bought new rulers. Which task has the highest cognitive demand?

- A. Find two different ways to show that two oddly shaped figures have equal area, and explain why each way works.
- B. Any area problem set in a real-world context.
- C. Memorize the area formulas for five shapes.
- D. Use $A = lw$ to find the areas of 20 rectangles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A teacher launches an open pattern task, but when students hesitate she lists the exact steps to reach the answer. What happens to the task's cognitive demand?

- A. It always rises when steps are provided.
- B. It declines to a procedure.
- C. It increases, because students get more guidance.
- D. It stays the same, because the task wording did not change.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

During a discussion in a portable classroom, a quiet student shares an idea about why two lines are parallel. The teacher wants another student to put that idea in their own words. The bell rings in ten minutes. Which move fits?

- A. "Great job!"
- B. "So you're saying the slopes are equal?"
- C. "Do you agree or disagree, and why?"
- D. "Who can say what she just said in your own words?"

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A teacher says, "So you're saying the slope stays the same because each step goes up 3?" Which talk move is this?

- A. Restating.
- B. Pressing for reasoning.
- C. Revoicing.
- D. Evaluating, since any teacher repeat judges the answer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Exit tickets on colored index cards show that 60% of students divided instead of multiplied when converting units. It was picture day, so the period was short. What should the teacher do next?

- A. Record the exit tickets as a test grade.
- B. Open the next lesson by analyzing that error, using unit reasoning to show when to multiply.
- C. Move on to the next lesson as planned.
- D. Reteach the whole unit from the beginning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Several ninth graders in a class that meets right after lunch cannot make sense of solving $2x + 3 = 11$ with symbols. The school recently repainted the hallways. Following the CRA sequence, what should the teacher do?

- A. Start with a formal proof of the properties of equality.
- B. Skip manipulatives, since they are only for elementary grades.
- C. Assign more symbolic equations for practice.
- D. Model the equation with algebra tiles or a balance, then drawings, then connect to the symbols.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher who won a district grant for classroom technology is designing a new unit on quadratic functions. The unit will start in February. Using backward design, what should she do first?

- A. Plan engaging activities with the new technology.
- B. Identify the standards and understandings students should reach.
- C. Pick the textbook chapter on quadratics.
- D. Write the unit test.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A teacher has set the unit's goals and just finished writing its performance task and rubric. In backward design, what comes next?

- A. Plan the learning experiences and instruction.
- B. Choose which textbook to adopt.
- C. Nothing; the unit is complete.
- D. Identify the desired results.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

In a class of 27 that meets in the science wing, a teacher lets students show their understanding of similar triangles through a written proof, a short video explanation, or a narrated dynamic-geometry construction. Which UDL principle is she applying?

- A. Multiple means of engagement.
- B. A modification that lowers the standard.
- C. Multiple means of representation.
- D. Multiple means of action and expression.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A teacher wants to apply multiple means of representation in a lesson on exponential growth. Which plan fits?

- A. Let students choose a poster, a podcast, or a test.
- B. Let students pick partners to raise motivation.
- C. Present growth with a table, a graph, a verbal story, and a folding-paper model.
- D. Present it once, clearly, because one representation is always enough.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

After a lesson on the distance formula, a teacher reflects. The principal stopped by for five minutes and smiled. Which reflection is most evidence-based?

- A. "18 of 27 exit tickets were correct; most errors came from subtracting negatives, so tomorrow opens with that."
- B. "Students seemed engaged, so the lesson worked."
- C. "Next time I will cover more content faster."
- D. "The principal liked it, so the lesson met its goal."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A teacher ~~planning on a Sunday evening~~ drafts goals for a lesson on slope. ~~Her district uses a new online gradebook.~~ Which is the best learning goal?

Target word: a well-formed learning goal **Decides:** The goal names the understanding, in a checkable form.

A	Adjacent category	Names an activity, not what students will learn.
B	Absolute language	Too broad to teach or check in one lesson.
C	Correct	Names the mathematics and how learning will be seen.
D	Confusion set	An engagement hope, not a learning outcome.

2. Answer A Concept 2

A teacher ~~with 110 students across four sections~~ is choosing an area task. ~~The department just bought new rulers.~~ Which task has the highest cognitive demand?

Target word: rank cognitive demand **Decides:** Students must explore, justify, and explain why.

A	Correct	Doing mathematics: no set path, and students must justify.
B	Absolute language	Context alone does not raise demand.
C	Inverted operation	Memorization, the lowest level.
D	Confusion set	Procedures without connections.

3. Answer B Concept 2 · transfer

A teacher launches an open pattern task, but when students hesitate she lists the exact steps to reach the answer. What happens to the task's cognitive demand?

Target word: effect of teacher moves on demand **Decides:** Giving the steps turns reasoning into a procedure.

A	Absolute language	Steps lower demand here.
B	Correct	Students now follow steps instead of reasoning.
C	Inverted operation	More direction means less thinking by students.
D	Dropped element	Demand depends on how the task is carried out, not just its wording.

4. Answer D Concept 3

During a discussion ~~in a portable classroom~~, a quiet student shares an idea about why two lines are parallel. The teacher wants another student to put that idea in their own words. ~~The bell rings in ten minutes.~~ Which move fits?

Target word: restating move **Decides:** A peer, not the teacher, rephrases.

A	Dropped element	Praise does not engage others with the idea.
B	Confusion set	That is revoicing: the teacher rephrases.
C	Adjacent category	Asks for evaluation, not a rephrasing.
D	Correct	Restating asks a peer to rephrase.

5. Answer C Concept 3 • transfer

A teacher says, “So you’re saying the slope stays the same because each step goes up 3?” Which talk move is this?

Target word: name the move **Decides:** The teacher rephrases and checks with the student.

A	Confusion set	Restating asks a peer to rephrase.
B	Adjacent category	Pressing asks why; this clarifies what.
C	Correct	The teacher restates the student’s idea and checks it.
D	Absolute language	Revoicing checks meaning without judging.

6. Answer B Concept 4

Exit tickets ~~on colored index cards~~ show that 60% of students divided instead of multiplied when converting units. ~~It was picture day, so the period was short.~~ What should the teacher do next?

Target word: use of formative evidence **Decides:** A shared error calls for a targeted response.

A	Confusion set	Uses formative evidence for a summative purpose.
B	Correct	Targets the specific misconception the data revealed.
C	Dropped element	Ignores the evidence that was collected.
D	Adjacent category	Overcorrects; the problem is one specific step.

7. Answer D Concept 5

Several ninth graders ~~in a class that meets right after lunch~~ cannot make sense of solving $2x + 3 = 11$ with symbols. ~~The school recently repainted the hallways.~~ Following the CRA sequence, what should the teacher do?

Target word: CRA application **Decides:** Go back to a concrete model, then connect to symbols.

A	Inverted operation	Moves further into the abstract, the reverse of CRA.
B	Absolute language	CRA supports learners at any age.
C	Dropped element	Stays at the abstract level that is not making sense.
D	Correct	Concrete to representational to abstract, with links at each step.

8. Answer B Concept 6

A teacher ~~who won a district grant for classroom technology~~ is designing a new unit on quadratic functions. ~~The unit will start in February.~~ Using backward design, what should she do first?

Target word: first stage of backward design **Decides:** Stage 1 is desired results.

A	Inverted operation	Activities are planned last.
B	Correct	Desired results come first; everything else aligns to them.
C	Adjacent category	A resource choice, not a desired result.
D	Confusion set	Assessment evidence is Stage 2.

9. Answer A Concept 6 • transfer

A teacher has set the unit's goals and just finished writing its performance task and rubric. In backward design, what comes next?

Target word: next stage **Decides:** Stages 1 and 2 are done.

A	Correct	Stage 3 follows the evidence stage.
B	Adjacent category	A resource decision, not a stage.
C	Dropped element	Stage 3 is still missing.
D	Inverted operation	Stage 1 is already done.

10. Answer D Concept 7

~~In a class of 27 that meets in the science wing,~~ a teacher lets students show their understanding of similar triangles through a written proof, a short video explanation, or a narrated dynamic-geometry construction. Which UDL principle is she applying?

Target word: UDL principle **Decides:** Choice in how students show learning.

A	Adjacent category	Engagement is about interest and motivation.
B	Inverted operation	Every option still has to show the same understanding.
C	Confusion set	Representation is how the teacher presents content.
D	Correct	Students choose how to demonstrate what they know.

11. Answer C Concept 7 • transfer

A teacher wants to apply multiple means of representation in a lesson on exponential growth. Which plan fits?

Target word: representation, not expression **Decides:** Present the content in several forms.

A	Confusion set	That is action and expression.
B	Adjacent category	That is engagement.
C	Correct	Several ways of perceiving and understanding the content.
D	Absolute language	UDL plans for varied learners.

12. Answer A Concept 8

After a lesson on the distance formula, a teacher reflects. ~~The principal stopped by for five minutes and smiled.~~ Which reflection is most evidence-based?

Target word: evidence-based reflection **Decides:** Uses student work against the goal and names a next step.

A	Correct	Specific data, a pattern, and a plan.
B	Adjacent category	An impression, not evidence of learning.
C	Dropped element	A change with no evidence behind it.
D	Camouflage	An observer's reaction is not student learning data.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Learning goals name the mathematics	A learning goal states what students will understand or be able to do, clearly enough to check, not the activity they will complete.	Writing the task (“complete the worksheet”) as the goal.	“Which is the best learning goal?”
2	Cognitive demand of tasks	Tasks range from memorization and procedures without connections (low) to procedures with connections and doing mathematics (high); teacher moves can lower demand during the lesson.	Judging a task’s demand by its context or length, or giving step-by-step hints that turn it into a procedure.	A set of tasks to rank, or a teacher who “helps” by giving the steps.
3	Productive talk moves	Revoicing (the teacher rephrases a student), restating (a peer rephrases), agree/disagree and why, adding on, wait time, and pressing for reasoning each do a different job.	Mixing up revoicing and restating, or treating praise as a talk move.	A quoted teacher prompt and “which move is this?”
4	Using evidence of student thinking	Formative evidence is useful only when it changes the next instructional step for the students who need it.	Collecting exit tickets and moving on as planned, or reteaching everything to everyone.	Exit-ticket or quiz results and “what next?”
5	Concrete-representational-abstract	When symbols are not making sense, move back to a concrete model, then a drawing, then symbols, connecting each step.	Assigning more symbolic practice, or believing manipulatives are only for young children.	Secondary students stuck at the symbolic level.
6	Backward design	Plan in three stages: identify desired results, decide the acceptable evidence, then plan learning activities.	Starting with favorite activities or the textbook chapter.	“What should the teacher do first when designing the unit?”
7	Universal Design for Learning	Plan from the start for multiple means of engagement (why), representation (what), and action and expression (how students show learning).	Mixing up representation (how content is presented) and expression (how students respond).	A teacher offering several ways to present or respond.
8	Evidence-based reflection	Reflect on a lesson by analyzing student work against the goal, then name a specific next step.	Judging a lesson by how it felt or by an observer’s comment.	Several reflection statements and “which is strongest?”

Course scope (public description): Methods of Teaching Secondary Mathematics is an in-depth exploration of secondary mathematics content and pedagogy, distinguishing between subject mastery and effective teaching practices. This course will prepare students to teach and reflect on secondary mathematics lessons that emphasize the use of effective teaching practices, including differentiation, questioning techniques, and curriculum development. The course also covers strategies for presenting mathematical material at the appropriate level, utilizing humanizing approaches, co-teaching, Multi-Tiered Systems of Support (MTSS), accommodations, and inclusion. This course does not have any required prerequisites.